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Solution. We are given $h(z)$ is nonzero on the boundary of the disc, which means that the quantity $c = \min_{|z|=1} |h(z)|$ satisfies $c > 0$. Let $M = \max_{|z|=1} |g(z)|$. then, if $M|\lambda| < c$ we have $|\lambda g(z)| \leq |h(z)|$ for all $z \in \partial D_1(0)$. By Rouché's theorem it follows that h and $h + \lambda g$ have the same number of zeroes inside $|z| < 1$. Now note that the condition $M|\lambda| < c$ is equivalent to $|\lambda| < \delta$, where $\delta = c/M$.

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Solution. $f(z)$ has a local inverse where $f'(z) \neq 0$. In this case, $f'(z) = 6z^2 + 6z - 12 = 6(z+2)(z-1)$. So $f'(z) \neq 0$ if $z \in \mathbb{C} \setminus \{-2, 1\}$.

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Solution. $f(z) = e^z$ gives $f'(z) = e^z$, so $f'(z) \neq 0$ for all $z \in \mathbb{C}$. Thus $f(z)$ has a local inverse at every point in $\mathbb{C}$. Suppose $e^z = w$, and f^{-1} is a local inverse of e^z defined at z . Then

$$(f^{-1})'(w) = \frac{1}{f'(z)} = \frac{1}{e^z} = \frac{1}{w}$$

Additional problem 1.

Solution. For this, we use the following fact from the proof of Rouché's Theorem:

$$\#\{\text{zeroes of } f \text{ inside } \gamma\} - \#\{\text{zeroes of } g \text{ inside } \gamma\} = \text{ind}_{h \circ \gamma}(0)$$

where $h(z) = \frac{f(z)}{g(z)}$. The condition $\text{Re}\left(\frac{f(z)}{g(z)}\right) > 0$ for $z \in \{\gamma\}$ says that the path $h \circ \gamma$ is contained in the half-space $\text{Re}(z) > 0$, which is a convex set that does not contain the point 0. Thus $\text{ind}_{h \circ \gamma}(0) = 0$.

Additional problem 2.

Solution. $f(z) = \frac{1}{1+z^2} = \frac{1}{(z-i)(z+i)} = \frac{-\frac{1}{2}i}{z-i} + \frac{\frac{1}{2}i}{z+i}$ has poles at $z = \pm i$, and $\text{Res}(f, i) = -\frac{1}{2}i$, $\text{Res}(f, -i) = \frac{1}{2}i$.

Suppose that γ_0 and γ_1 are two paths in $\mathbb{C} \setminus [-i, i]$ that both start at z_0 and end at z_1 . Consider the closed path $\gamma = \gamma_1 - \gamma_0$. Since $\{\gamma\} \subset \mathbb{C} \setminus [-i, i]$ then f has no poles on $\{\gamma\}$, so

$$\begin{aligned} \int_{\gamma} \frac{1}{1+z^2} dz &= 2\pi i \left(-\frac{i}{2} \text{ind}_{\gamma}(i) + \frac{i}{2} \text{ind}_{\gamma}(-i) \right) \\ &= \pi \left(\text{ind}_{\gamma}(i) - \text{ind}_{\gamma}(-i) \right) \end{aligned}$$

Since $[-i, i] \subset \mathbb{C} \setminus \{\gamma\}$, it follows that we can connect $-i$ to i by a path that does not intersect γ . Hence $\text{ind}_{\gamma}(i) = \text{ind}_{\gamma}(-i)$. We conclude that $\int_{\gamma} f(z) dz = 0$ for all closed paths γ contained in $\mathbb{C} \setminus [-i, i]$. Thus,

$$\int_{\gamma_1 - \gamma_0} \frac{1}{1+z^2} dz = \int_{\gamma_1} \frac{1}{1+z^2} dz - \int_{\gamma_0} \frac{1}{1+z^2} dz = 0.$$